

MAT 548: Geometry and Topology I

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1 August 26

Definition 1.1 (Topological manifold) — An n -dimensional topological manifold is a second-countable Hausdorff space X that is locally Euclidean, i.e. for all $x \in X$, there is an open $U \subseteq X$ containing x that is homeomorphic to an open subset of $\mathbb{R}^n$.

Note that this implies that manifolds are locally compact because open sets in $\mathbb{R}^n$ are locally compact. Topological manifolds also satisfy a useful property called paracompactness:

Definition 1.2 (Paracompactness) — A topological space X is paracompact if any open cover $\{U_\alpha\}_{\alpha \in A}$ of X has a locally finite refinement; this means that there is an open cover $\{V_\beta\}_{\beta \in B}$ of X such that

- (1) for all $\beta \in B$, there exists $\alpha \in A$ satisfying $V_\beta \subseteq U_\alpha$.
- (2) for all $x \in X$, there is a neighborhood U of x such that U intersects finitely many elements of $\{V_\beta\}_{\beta \in B}$.

Proposition 1.3. Topological manifolds are paracompact. More strongly, given a topological manifold X , an open cover $\{U_\alpha\}_{\alpha \in A}$, and any basis $\mathcal{B}$ for the topology on X , there exists a countable, locally finite refinement of $\{U_\alpha\}_{\alpha \in A}$ consisting of elements of $\mathcal{B}$.

Proof. Let $K_1, K_2, K_3, \dots$ be a set of compact subsets of X such that $K_i \subseteq \text{Int } K_{i+1}$ and $\bigcup_{i=1}^\infty K_i = X$. Then, define the open set $O_i = \text{Int } K_{i+2} \setminus K_{i-1}$ and the compact set $C_i = K_{i+1} \setminus \text{Int } K_i$. (Such a set exists in general for any second-countable locally compact Hausdorff space; see Prop. A.60 in Lee.)

Now, for each $x \in C_i$, there exists a U_α such that $x \in U_\alpha$ and a basis element B_x containing x such that $B_x \subseteq U_\alpha \cap O_i$. Since C_i is compact, we can find a finite subcover of the B_x , say $\{B_{i,1}, B_{i,2}, \dots, B_{i,N_i}\}$. The set $\{B_{i,j} : j \leq N_i\}$ is clearly a refinement of $\{U_\alpha\}_{\alpha \in A}$, and we will show local finiteness.

To do this, given $x \in X$, pick k such that $x \in O_k$. If $B_{m,*} \cap O_k \neq \emptyset$, then $C_m \cap O_k \neq \emptyset$, which implies that $k-1 \leq m \leq k+2$. There are only finitely many $B_{m,*}$ satisfying this, so O_k intersects finitely many sets in our refinement, proving that the refinement is locally finite and thus X is paracompact. $\square$

Proposition 1.4. Let X be a topological manifold and $\{U_\alpha\}_{\alpha \in A}$ an open cover that is locally finite. Then, there exists a locally finite open cover $\{V_\alpha\}_{\alpha \in A}$ such that for all $\alpha \in A$, $\overline{V_\alpha} \subseteq U_\alpha$.

Proof.

$\square$

Definition 1.5 (C^∞ manifold) — Let X be an n -dimensional topological manifold. An **atlas** on X is an open cover $\{U_\alpha\}_{\alpha \in A}$ of X with homeomorphisms

$$h_\alpha: U_\alpha \rightarrow O_\alpha \overset{\text{open}}{\subseteq} \mathbb{R}^n$$

todo; didn't understand the one shown in class

such that for all $\alpha, \beta \in A$ with $h_\beta \circ h_\alpha^{-1}$ is a diffeomorphism between $h_\alpha(U_\alpha \cap U_\beta)$ and $h_\beta(U_\alpha \cap U_\beta)$. The pairs (U_α, h_α) are called **coordinate charts** of the underlying topological manifold and the composed maps $h_\beta \circ h_\alpha^{-1}$ are called **transition maps**. Then, the pair $(X, \{(U_\alpha, h_\alpha)\}_{\alpha \in A})$, abbreviated as $(X, \{U_\alpha\}_{\alpha \in A})$, is a C^∞ **manifold** or **smooth manifold** (which we will just shorten to “manifold”).

The transition maps should be thought of as a “local change of coordinates.”

Definition 1.6 (Smooth map to $\mathbb{R}^N$) — Let $(X, \{U_\alpha\}_{\alpha \in A})$ be a manifold and $f: X \rightarrow \mathbb{R}^N$ a function. We say that f is (infinitely) differentiable (or smooth, or C^∞) if for all $\alpha \in A$, $f \circ h_\alpha^{-1}: h_\alpha(U_\alpha) \rightarrow \mathbb{R}^N$ is C^∞ .

Definition 1.7 (Smooth map between manifolds) — Let $(X, \{U_\alpha\}_{\alpha \in A})$ and $(Y, \{V_\beta\}_{\beta \in B})$ be manifolds. A map $F: X \rightarrow Y$ is differentiable (or smooth or C^∞) if for all $x \in X$, there exists $U_\alpha \ni x$ and $V_\beta \ni f(x)$ and an open set O around $h_\alpha(x)$ such that $F(h_\alpha^{-1}(O)) \subseteq V_\beta$, and the composition

$$O \xrightarrow{h_\alpha^{-1}} U_\alpha \xrightarrow{F} V_\beta \xrightarrow{h_\beta} h_\beta(V_\beta)$$

is smooth.

Definition 1.8 (Compatibility of atlases) — Two atlases $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ and $\mathcal{V} = \{V_\beta\}_{\beta \in B}$ on X are **compatible** if $\mathcal{U} \cup \mathcal{V}$ is an atlas (so transition maps between elements of $\mathcal{U}$ and $\mathcal{V}$ are smooth).

Given an open subset W of a manifold X and a diffeomorphism $\psi: W \rightarrow O \stackrel{\text{open}}{\subseteq} \mathbb{R}^n$, the atlas of X is compatible with the atlas resulting from adding (W, ψ) to the atlas. This means that given an atlas $\mathcal{U}$ of X , we can define

$$\mathcal{U}_{\max} = \mathcal{U} \cup \{(W, \psi) : \psi \text{ is a diffeomorphism from } W \stackrel{\text{open}}{\subseteq} X \text{ to an open set in } \mathbb{R}^n\}.$$

It is not hard to check that this is indeed an atlas, and $\mathcal{U}_{\max}$ is maximal with respect to inclusion, meaning it is not properly contained by any other atlases on X .

Example 1.9

Some examples of manifolds:

- Any open subset $U \subseteq \mathbb{R}^n$ can be made a manifold with the atlas $\{(U, \text{id})\}$. (What is the corresponding maximal atlas then?)
- The general linear group $GL_n(\mathbb{R})$, which is open in $\mathbb{R}^{n^2}$, can be made a manifold as stated above. $GL_n(\mathbb{R})$ is also of course a group where the multiplication map $(A, B) \mapsto AB$ and inverse map $A \mapsto A^{-1}$ are both C^∞ . This is an example of a **Lie group**.

2 August 28

2.1 Some more examples of manifolds

The sphere $S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$ is a manifold of dimension of n with charts

$$\begin{aligned}\varphi_k^\pm : B_1^n(0) &\rightarrow S^n \\ x &\mapsto (x_1, \dots, x_{k-1}, \sqrt{1 - \|x\|^2}, x_k, \dots, x_n)\end{aligned}$$

where $B_r^n(y)$ indicates an open n -dimensional ball centered at y with radius r and $k = 1, \dots, n$. (It is very simple to verify that the transition maps are diffeomorphisms; try it!)

The sphere can be acted upon by $\mathbb{Z}_2$ via the antipodal map $x \mapsto -x$ (more precisely, identity element goes to identity homeomorphism and the other element goes to the antipodal homeomorphism), and the orbit space $S^n/\mathbb{Z}_2$ is what we call **real projective space**, denoted $\mathbb{RP}^n$; equivalently, this is the set of one-dimensional subspaces of $\mathbb{R}^{n+1}$ (since we can identify each subspace with a pair of antipodal points on S^n). This is a manifold of dimension n with the coordinate charts

$$\begin{aligned}\varphi_k : \mathbb{R}^n &\rightarrow \mathbb{RP}^n \\ (x_1, \dots, x_{k-1}, x_{k+1}, \dots, x_{n+1}) &\mapsto (x_1, \dots, x_{k-1}, 1, x_{k+1}, \dots, x_{n+1})\end{aligned}$$

for $k = 1, \dots, n+1$. We need to check that the transition maps are diffeomorphisms in this case. Note that $\varphi_k(\mathbb{R}^n)$ is the subset of $\mathbb{RP}^n$ whose k th coordinate is nonzero, so if $j \neq k$, then $\varphi_j(\mathbb{R}^n) \cap \varphi_k(\mathbb{R}^n)$ is the subset whose j th and k th coordinates are both nonzero, so the map $\varphi_j \circ \varphi_k^{-1} : \varphi_k(\mathbb{R}^n) \rightarrow \varphi_j(\mathbb{R}^n)$, which is simply the division-by- x_j map

$$(x_1, \dots, 1, \dots, x_k, \dots, x_n) \mapsto \left(\frac{x_1}{x_k}, \dots, \frac{1}{x_k}, \dots, 1, \dots, \frac{x_n}{x_k} \right),$$

is differentiable because $x_k \neq 0$, as is its inverse because $x_j \neq 0$.

A generalization of projective space is the **Grassmannian** $\text{Gr}(k, n)$ for $0 \leq k \leq n$, which is the space of all k -dimensional subspaces of $\mathbb{R}^n$.

One last example is given by taking a function $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ and looking at the pre-image $M = F^{-1}(0)$ (0 is of course arbitrary). Of course, this need not be a manifold, such as in the case $F = x^2 + y^2 - z^2$ (no chart exists around $(0, 0, 0)$); on the other hand, the simple condition that the derivative $(\nabla F)_x$ at any $x \in M$ is nonzero is enough to make M a manifold!

2.2 Partitions of unity

The idea of partitions of unity is to write a smooth function $f : X \rightarrow \mathbb{R}$ as a sum of possibly infinitely many smooth functions such that the sum locally looks like a finite sum around any $x \in X$, so we are not restricted to finite sums while also not having to worry about issues of convergence.

The first step of this is to construct a smooth function $\mathbb{R}^n \rightarrow \mathbb{R}$ that is supported on a closed ball.

i don't get how he computed the dim as $k(n-k)$

prove this

bump f'ns

3 September 2

partitions of
unity stuff

def mflid w
bd

The geometric idea of a tangent vector is that we want to take a C^∞ curve $\gamma: (-\varepsilon, \varepsilon) \rightarrow X$, and then the tangent vector to γ at $\gamma(0)$ is $\frac{d\gamma}{dt}(0)$. This, however, depends on the choice of coordinates. We will instead use a coordinate-free, more abstract path to define tangent vectors.

Definition 3.1 (Germs of a manifold) — Let X be a manifold. Consider the collection of pairs

$$C_p = \{(U, f) : p \in U \overset{\text{open}}{\subseteq} X, f \in C^\infty(U)\}$$

and define an equivalence relation $\sim$ on C_p by $(U, f) \sim (U', f')$ if there is a neighborhood W of p contained in $U \cap U'$ such that $f \equiv f'$ on W . The **germs** at p are the equivalence classes, the collection of which we denote by $C_p^\infty(X)$.

The germs at p form a unital associative algebra where the operations of addition, multiplication, and scalar multiplication are given by

$$\begin{aligned} [(U, f)] + [(V, g)] &= [(U \cap V, f + g)] \\ [(U, f)][(V, g)] &= [(U \cap V, fg)] \\ \lambda[(U, f)] &= [(U, \lambda f)] \end{aligned}$$

where $[(W, h)]$ is the germ containing h .

Exercise 3.2. Check that these operations are well-defined.

The upshot is that we can understand local properties of a function f just by looking at its equivalence class, somehow?

Definition 3.3 (Tangent vector) — A **tangent vector** v at $p \in X$ is a linear map $V: C_p^\infty(X) \rightarrow \mathbb{R}$ such that

$$v(fg) = v(f)g(p) + f(p)v(g)$$

im too tired
for this shit

4 September 4

there was
some stuff
with showing
that partials
make
basis of tan-
gent space

Definition 4.1 (Tangent bundle) — The tangent bundle TX is the disjoint union

$$\coprod_{x \in X} T_x X$$

equipped with a map $\pi: TX \rightarrow X$ given by $\pi(T_x X) = x$. A **section** of TX over an open subset $U \subseteq X$ is a map $\sigma: U \rightarrow TX$ such that $\pi \circ \sigma = \text{Id}_U$, i.e. $\sigma(x) \in T_x(X)$ for all $x \in U$.

If $\varphi: U \rightarrow \mathcal{O} \subseteq \mathbb{R}^n$ gives local coordinates on X , then

$$\frac{\partial}{\partial x_j}: \mathcal{O} \rightarrow T\mathcal{O}$$

are local sections of $T\mathcal{O}$ and we have a bijection $\mathcal{O} \times \mathbb{R}^n \rightarrow T\mathcal{O}$ by taking

$$(x, c_1, \dots, c_n) \mapsto \sum_{j=1}^n c_j \frac{\partial}{\partial x_j} \Big|_x.$$

The inverse $T\mathcal{O} \rightarrow \mathcal{O} \times \mathbb{R}^n$ induces local coordinates on $T\mathcal{O}$, hence on TX .

Proposition 4.2. If X is a manifold then TX is a C^∞ manifold.

Proof. We need to check that the transition maps between the charts described above are diffeomorphisms. □

do that.

Definition 4.3 (Tangent vector field) — A **tangent vector field** v on an open subset $U \subseteq X$ is a C^∞ section $\sigma: U \rightarrow TU$.

5 September 9